Numerical Methods

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NUMERICAL METHODS

1. ROOT FINDING
Bisection: halve interval

Secant method
Fixed point: x = g(x)

2. INTERPOLATION
Lagrange polynomial
Newton divided differences
Cubic splines

3. NUMERICAL INTEGRATION
Trapezoidal rule
Simpson’s 1/3 rule
Simpson’s 3/8 rule
Gauss quadrature

4. ODE SOLVERS
Euler method
Runge-Kutta (RK4)
Adams-Bashforth

5. LINEAR SYSTEMS
Gaussian elimination
LU decomposition
Jacobi iteration
Gauss-Seidel